

Power Module for BLDC and Servo Motor

FEATURES

- Power Module for BLDC and SERVO Motor
- Input Voltage Range: DC 24V to 36V
- 4ch pass-through channels ($V_{out} = V_{in}$) for direct motor power delivery
- 2ch switching regulator channels with fixed 12V output, up to 2.0A each, also usable as motor power
- Output Voltage Spec:
 - 4ch 24V for Motor
 - 2ch 12V for Motor or Other loads if needed
- Output Current Spec:
 - 4ch 24V – Depends on Input Current Capability
 - 2ch 12V – Up to 2.0A

APPLICATIONS

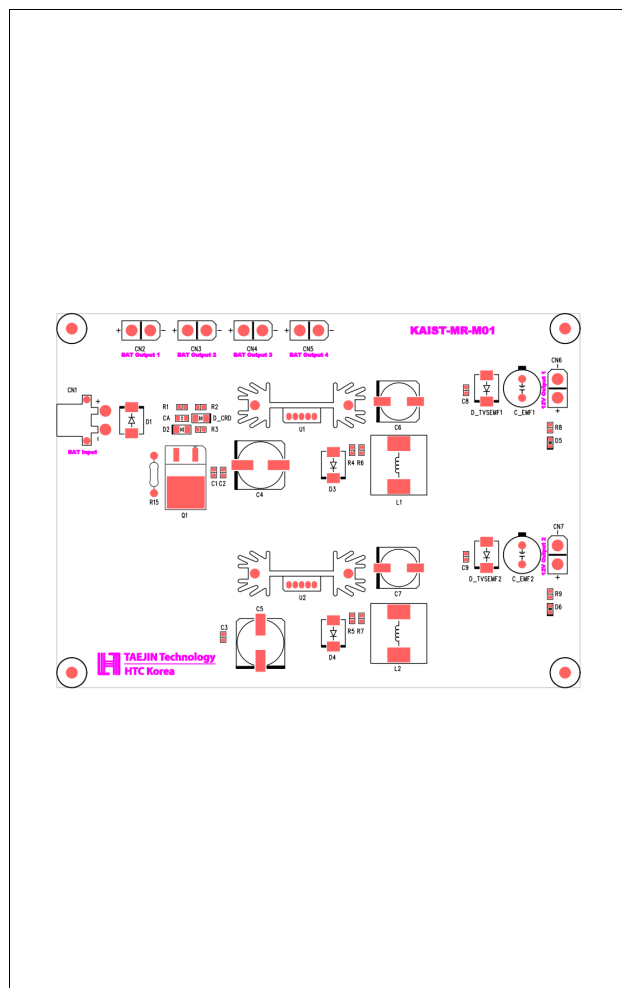
- Drones / Robotics
- Autonomous Ground Robots

DESCRIPTION

This power distribution module is designed for BLDC and Servo motor drive systems, accepting a wide input voltage range of 24V to 36V. It provides six independent output channels for powering motors: four pass-through channels (CH1–CH4) that deliver the input voltage directly, and two switching regulator channels (CH5–CH6) that step down the input to a regulated 12V, each capable of supplying up to 2.0A. All six channels are fundamentally intended as motor power outputs, though they can also be used to power other loads if needed.

APPLICATION GUIDE

This module is primarily intended to supply power to multiple BLDC or servo motors within a single system, with six independent channels available for that purpose. The four pass-through channels deliver the input voltage directly and are suited to motors or drive stages that operate across the full 24V–36V range. The two 12V/2.0A switching regulator channels provide a regulated motor power option where a lower, stable voltage is required. While each channel is fundamentally designed for motor power, they can also be used to supply other loads in the system if the application calls for it.

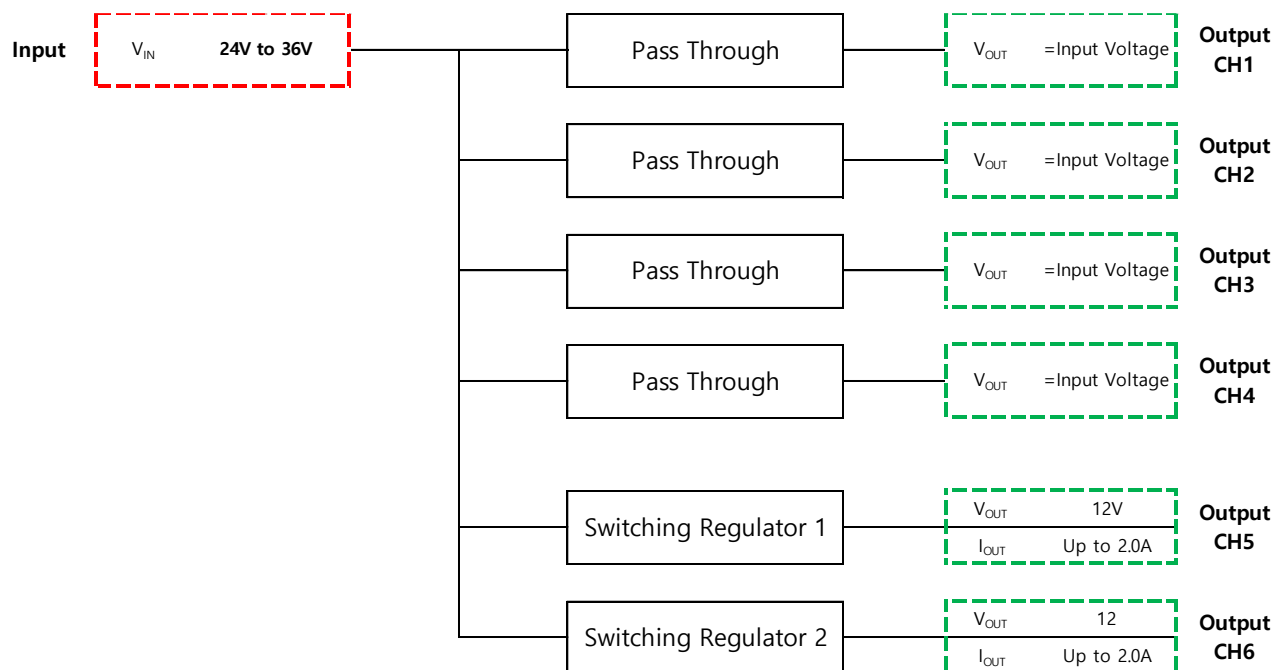


ORDERING INFORMATION

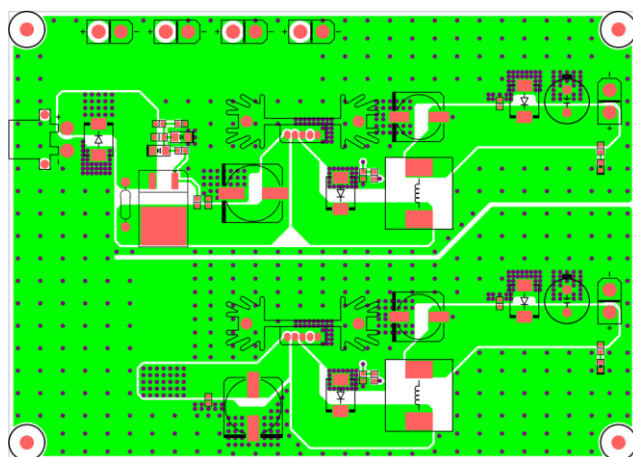
Part Number	KAIST-MR-M01
Device	Power Module for BLDC and Servo Motor
Package	Assembled Module (10cm X 14cm)

Power Module for BLDC and Servo Motor

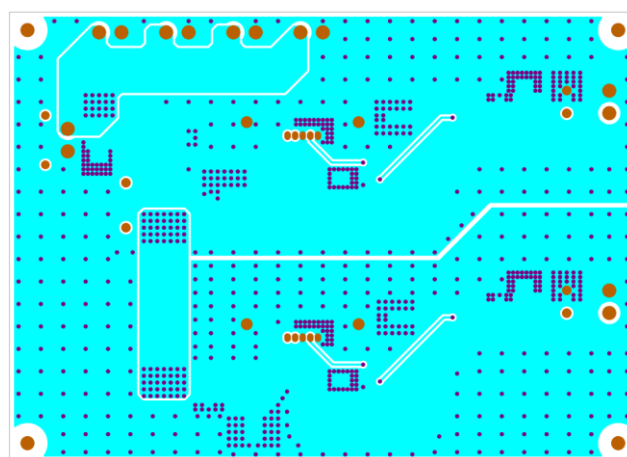
POWER TREE



LAYOUT AND SIZE INFORMATION



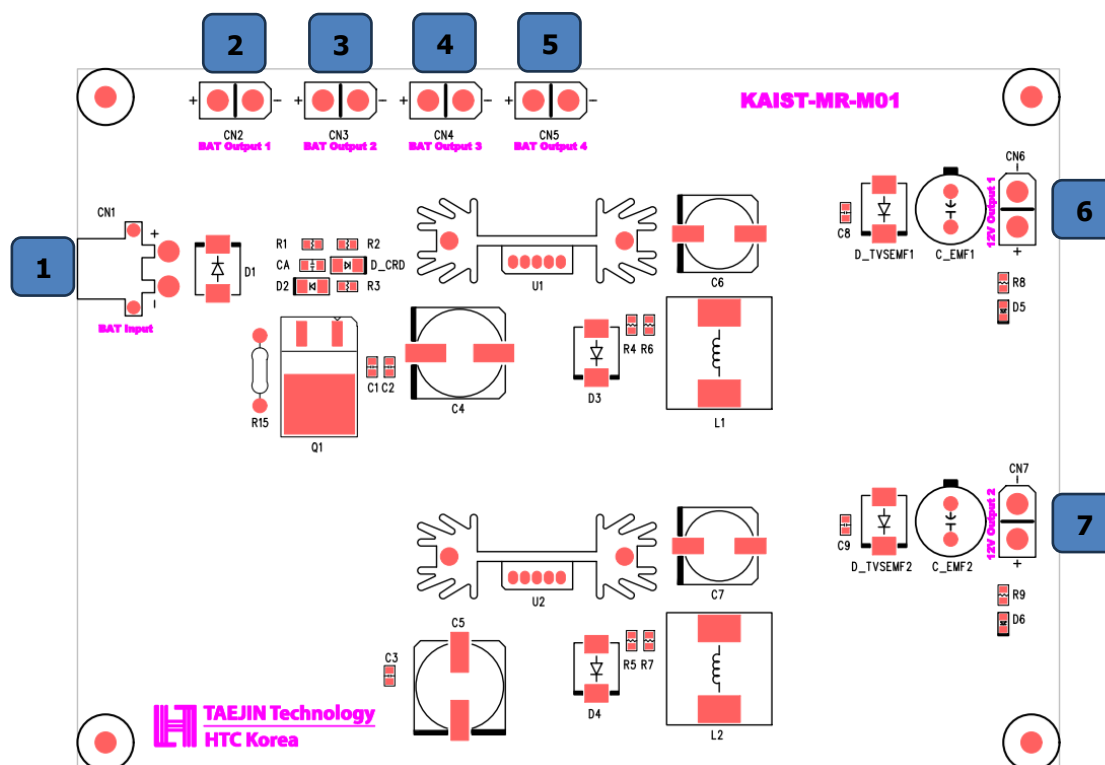
TOP



BOTTOM

Power Module for BLDC and Servo Motor

I/O PORT CONFIGURATION AND DESCRIPTION



I/O PORT CONFIGURATION AND DESCRIPTION

Designator	Port Name	Port Description
1	Input Connector (XT-30 Female Connector)	Input Connector : DC 24V to 36V
2	Output Connector - CH1 (XT-30 Female Connector)	Output Connector - CH1: =Input Voltage (Depends on Input Current Capability)
3	Output Connector - CH2 (XT-30 Female Connector)	Output Connector - CH2: =Input Voltage (Depends on Input Current Capability)
4	Output Connector - CH3 (XT-30 Female Connector)	Output Connector – CH3: =Input Voltage (Depends on Input Current Capability)
5	Output Connector - CH4 (XT-30 Female Connector)	Output Connector – CH4: =Input Voltage (Depends on Input Current Capability)
6	Output Connector - CH5. (XT-30 Female Connector)	Output Connector - CH5: 12V / Up to 2.0A
7	Output Connector - CH6 (XT-30 Female Connector)	Output Connector - CH6: 12V / Up to 2.0A

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REVISION NOTICE

The description in this datasheet is subject to change without any notice to describe its electrical characteristics properly.